

Marllon Rodrigues

Maritime Systems & Software Engineer

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Profile

Engineer at the intersection of **maritime operations and digital systems**. Built and shipped a Port Identity Layer for international shipping at a Singapore maritime SaaS startup — covering vessel onboarding, bunkering workflows, and digital compliance. Deployed on-site across two Petrobras FPSOs (P80/P83) in Singapore: 50,000+ tags, 1,700+ field devices, 7,000+ IEC 61850 signals across 105 IEDs, live data collection via Modbus TCP and industrial protocols. Experienced in structuring industrial data for dashboards, APIs, and cloud-connected systems. **English C1**.

Technical Skills

Industrial Protocols & Data: Modbus TCP, IEC 61850 MMS, Profinet, Ethernet/IP, OPC; AVEVA Historian (time-series data); SCADA tag configuration (50,000+ tags); IT/OT integration; industrial network architecture.

Software & Cloud: TypeScript, Python, Node.js, PostgreSQL, REST APIs, Docker, GitHub Actions CI/CD, Git, Prisma, Firestore; experience with cloud-connected data pipelines.

Maritime & Offshore Systems: AVEVA Plant SCADA (Citect), System Platform, InTouch, Power Operation (EPO), MES; SEL Relays (F650, 869, 845); Schneider M580 PLCs; Siemens S7-1200/1500; FPSO systems; Danfoss VFDs.

Experience

Automation Analyst, Tyconnex (*Singapore/On-site*)

Feb 2026 – Present

- **Digital Systems on Offshore Platforms (Petrobras P80/P83):** On-site engineer for AVEVA Plant SCADA on two FPSOs — 50,000+ tags, 1,700+ field devices, Schneider M580 PLCs, 168 panels across 5 industrial subnets. Responsible for data collection configuration, tag mapping, and live system validation.
- **IEC 61850 & Protocol Integration:** Configured SEL protection relays (F650, 869, 845) via IEC 61850 MMS — 7,000+ signals catalogued across 105 IEDs. Commissioned gas compressors via Modbus TCP over RTDS; resolved critical data scaling fault affecting process values.
- **VFD Data Integration:** Commissioned and validated Danfoss VFDs (02A/02B/03A/03B) via Modbus TCP; diagnosed W1007/W1022 faults through live register-level analysis of industrial data streams.
- **Network & Device Audit:** Physical and logical survey of 77 switch actions across Hirschmann industrial switches; reconciled 68 discrepancies including rogue devices across 5 subnets (TES001–TES027).
- **TagFinder Tool (Offline Data App):** Built a field tool (single HTML, 4.58MB) consolidating 3,432 P80 + 1,785 P83 equipment records, Modbus address tables, IEC 61850 signal catalog, and IP mapping for 1,339 devices — used daily by the team as a real-time data reference.

Full Stack Engineer, Bunkerchain (*Singapore/Remote*)

Jun 2025 – Feb 2026

- **Maritime Digital Platform (MVP 3.0):** Full-stack development on Marine Vessel Pass — a Port Identity Layer for international shipping. FSM-driven port call lifecycle covering vessel onboarding, bunkering, and regulatory compliance workflows. Worked directly with the Singapore founding team.
- **API Design & Data Architecture:** Designed REST APIs for vessel identity, port calls, and fuel sample traceability; PostgreSQL schema design for multi-signatory workflows; real-time data sync across web and mobile clients.
- **Cryptographic Trust System:** Built VPS seal verification — ECDSA digital signatures, public key registry, seal lifecycle state machine; ensures data integrity for physical fuel samples across the supply chain.
- **Security & Audit Logging:** Closed 10+ cross-tenant authorization vulnerabilities; implemented hash-chained audit logs with global sequence integrity for regulatory traceability.
- **eBDN Platform (Methanol):** End-to-end delivery of the Terminal Role on the electronic Bunker Delivery Note system — architecture to live client demo approval.

Systems Developer, MAIMA Technology

Aug 2024 – May 2025

- **Process Automation & Data Pipelines:** Python robots automating 6 critical daily SAP processes, reducing manual execution time from 8 hours to under 2 minutes; distributed microservices architecture (Image Analysis, Data, BFF layers).
- **Predictive Analytics:** C# system for machine downtime prediction based on product quality data analysis; AI-powered CI/CD pipelines with automated code review and security scanning.

Automation Technician, S.Build Automation

Jun 2022 – Aug 2024

- **SCADA & Historian:** AVEVA System Platform, InTouch HMI, and Historian configuration in critical industrial environments; Python scripts for mass SQL data updates in AVEVA MES.
- **Industrial Networks:** Profinet, Ethernet/IP, Modbus, OPC across Siemens (S7-1200/1500), Rockwell, and Schneider PLC projects; upgrade of 20+ servers via ThinManager.

Certifications

- **CSI AVEVA** Manufacturing Execution System (MES) 2020 – Operations & Performance Exams
- **CSI AVEVA** Application Server 2023 Exam
- **.NET Developer Bootcamp** (DIO – 93h) **Node.js Fundamentals** (DIO – 30h)

Education

UNINTER, Systems Analysis and Development	2023 – 2025
SENAI Roberto Simonsen, Computer Networks Technician	2021 – 2022
SENAI Roberto Simonsen, Electronics Maintenance Electrician	2020 – 2022

Languages

- English (C1 – Advanced)
- Portuguese (Native)